PUNEET CHANDNA

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Education

Bachelor of Technology (B.Tech) in Computer Science and Engineering (CSE)
Vellore Institute of Technology, Chennai — *Expected Graduation: 2026*

Experience

Research Intern — May 2025 – July 2025
Centre for e-Automation Technologies (CeAT), VIT Chennai
- Developed a research framework built on **CloudSim Plus** to analyze and evaluate the performance of the **Hippopotamus Optimization (HO)** algorithm for efficient **Virtual Machine (VM) placement** in cloud data centers.
- Enhanced experimental scalability and reproducibility by modularizing simulation components, enabling testing across multiple datacenter configurations.
- Authored a research paper demonstrating simulation results and explaining how researchers can modify and optimize parameters within the developed **CloudSim Plus–based framework** to observe significant variations in VM placement efficiency and energy-performance metrics — GitHub: Repo

Full Stack Developer Intern — Dec 2024 – Feb 2025
Daira Edtech Pvt Limited
- Contributed to the robust backend infrastructure by developing features and RESTful APIs for the Vidhira EdTech platform.
- Ensured data structure integrity and scalability leading to 30% improvement in data retrieval efficiency by designing effective data models to support new features on the Vidhira platform.

Web Development Intern — Sept 2024 - Oct 2024
Indian Institute of Technology Bombay (IIT Bombay)
- Significantly reduced data payload size by 90% for low-bandwidth environments by implementing numerical status codes in JSON responses instead of full text messages.
- Reduced system latency and improved data retrieval speed by optimizing backend database queries for the SaaS platform.
- Enhanced backend robustness and API reliability by implementing Joi validation for data schemas within the SaaS project.

Projects

BookMyEvent — Jan 2024 – Mar 2024
- Developed an event management application capable of processing around 10,000 ticket authentications per event, streamlining operations by integrating real-time QR code scanning using an ESP32 kit.

Technologies Used: Node.js, Express.js, Computer Vision, ESP32, Flutter, MongoDB — GitHub: Server — App

Water Brakes — Aug 2024 - Nov 2024
- Developed a streamlit application assisting farmers in optimizing water management by analyzing contour map data to recommend precise swale and trench locations.
- Enabled data-driven decisions for improved groundwater recharge and flood prevention by processing UTM data from CSV files and generating interactive visualizations using Plotly and Matplotlib.

Technologies Used: Streamlit, Python, Plotly, Matplotlib — GitHub: Repo

High-Performance Parallel AES Encryption Framework — Jan 2025 - Apr 2025
- Engineered a high-performance AES encryption system that reduced encryption time by 51% on large payloads through parallel processing using Python's ProcessPoolExecutor and key-dependent dynamic S-boxes generated via SHA-512.
- Enhanced data throughput by 21.9% (up to 362 KB/sec) compared to standard AES by designing a multi-core CTR-mode encryption pipeline for distributed processing.
- Achieved a 49.95% avalanche effect (near-ideal 50%), confirming cryptographic robustness and validating the security of the dynamic S-box approach.

Technologies Used: Python, Multiprocessing, SHA-512, Matplotlib, NumPy

Emotion-Aware Movie Recommendation System — Jan 2024 - May 2025
- Built an NLP-powered movie recommendation engine capable of interpreting hybrid emotional states by leveraging SBERT embeddings to map user text to nuanced mood categories.
- Achieved a 0.71 Macro F1 score by designing a custom classification approach combining cosine similarity with sentiment polarity, outperforming standard random forest baselines.
- Developed a complete text-processing pipeline that transformed raw user input into enriched emotional features using PCA for effective dimensionality reduction.

Technologies Used: Python, SBERT, spaCy, Machine Learning, Pandas — GitHub: Repo

Interpretable Hybrid Ensemble Model for Crop Stress Detection — Aug 2025 - Nov 2025
- Built a hybrid machine learning pipeline for binary plant-health classification using 30-day physiological sensor data, enabling early and reliable stress detection.
- Combined an LSTM network with attention for capturing temporal patterns with an XGBoost model in a weighted ensemble to improve interpretability and predictive stability.
- Evaluated the system using 8-timestep sliding windows, demonstrating adaptability of the architecture and measurable improvements in early-stage stress detection metrics.

Technologies Used: Python, LSTM, XGBoost, Scikit-learn, Pandas, Time-Series Analysis

ASCII Video Insanity (Terminal-Based Media Player) Jun 2025 - Jul 2025

- Built a high-performance CLI media player capable of rendering videos as colorized ASCII art at 30+ FPS by optimizing frame-buffer handling and stdout flushing.
- Designed real-time image processing pipelines with OpenCV to resize, map, and convert pixel data into ANSI-colored character streams with minimal latency.
- Created a custom RGB–density mapping algorithm that converts pixel intensity into ASCII characters while preserving visual fidelity and aspect ratio across terminal sizes.

Technologies Used: Python, OpenCV, PIL, ANSI Escape Codes — GitHub: Repo

Skills

- **Programming Languages:** Python, C++, C, Java, HTML, CSS, SQL, JavaScript, Bash.
- **Databases & Cloud:** MongoDB, PostgreSQL, Amazon Web Services (AWS), Google Cloud (GCP), Docker.
- **Tools & Frameworks:** Git, React, Next.js, Node.js, Express.js, RESTful APIs, Linux.
- **Core Concepts:** Data Structures & Algorithms, DBMS, CN, OS, Cryptography (SHA-512/AES), Parallel Processing (Multiprocessing),System Design, CI/CD, System Optimization.

Positions of Responsibility

- **Outreach Lead, DAO Community:**
  - Drove 50% community membership growth within 6 months by leading strategic outreach initiatives and enhancing engagement through numerous on-campus workshops and sessions facilitated via collaborations with leading Web3 companies (including Spheron, Polygon, blockOn, Huddle01, zkwallet).
  - Established the DAO as a key partner in the national/global Web3 ecosystem by securing alliances for 15+ nationwide hackathons and the TOKEN2049 Singapore conference, and playing a key organizing role in 5 major hackathons (250 participants each); notably led a 35-member team for Defy24's record-breaking sponsorship and profit.
- **Blockchain Lead, Hackclub:**
  - Expanded the blockchain department by 300% in 6 months by recruiting and actively mentoring over 25 junior students.
  - Developed members' technical expertise by designing and delivering online blockchain workshops, creating learning roadmaps, and providing personalized skill development guidance.